

# Bioinformatics Practical 3

Name: Sidharthan S C

Roll No: BE21B039

## Question 1

The image shows two screenshots of the UniProt database interface. The top screenshot displays the 'Sequence' tab for a protein entry. It includes a sidebar with navigation options like Function, Names & Taxonomy, Subcellular Location, Disease & Variants, PTM/Processing, Expression, Interaction, Structure, Family & Domains, Sequence, and Similar Proteins. The main content area shows the 'Sequence' tab selected, with a 'Sequence status' of 'Complete'. It also provides a 'See also' link to UniParc or UniRef. The sequence itself is displayed in a table format with row numbers and the amino acid sequence. The bottom screenshot shows the 'Function' tab for the same protein entry. It includes a sidebar with navigation options like Function, Names & Taxonomy, Subcellular Location, Disease & Variants, PTM/Processing, Expression, Interaction, Structure, Family & Domains, Sequence, and Similar Proteins. The main content area shows the 'Function' tab selected, with a 'Function' section describing the protein's role in the mitochondrial outer membrane and apoptosis. It also includes an 'Activity regulation' section and a 'Features' section.

**UniProt** BLAST Align Peptide search ID mapping SPARQL UniProtKB Advanced | List Search

Function Variant viewer 100 Feature viewer Genomic coordinates Publications External links History

### Sequence<sup>i</sup>

Sequence status<sup>i</sup> Complete

See also sequence in UniParc or sequence clusters in UniRef

Tools Download Add Highlight Copy sequence

Length 283  
Mass (Da) 30,773

Last updated 2007-01-23 v2  
Checksum<sup>i</sup> 89BA3378B04020D5

|     |             |     |            |     |            |     |            |     |            |     |            |     |            |     |            |     |            |     |            |     |            |
|-----|-------------|-----|------------|-----|------------|-----|------------|-----|------------|-----|------------|-----|------------|-----|------------|-----|------------|-----|------------|-----|------------|
| 10  | MAVPPPTAYDL | 20  | GKSARDVFTK | 30  | GYGFGLIKLD | 40  | LKTKSENGLE | 50  | FTSSGSANTE | 60  | TTKVTGSLET | 70  | KYRWTEYGLT | 80  | FTEKNWTDNT | 90  | LGTEITVEDQ | 100 | LARGKLKTFD | 110 | SSFSPNTGKK |
| 120 | NAKIKTGYKR  | 130 | EHINLGCDD  | 140 | FDIAGPSIRG | 150 | ALVLGYEGWL | 160 | AGYQWNFETA | 170 | KSRVTQSNFA | 180 | VGKTDQFQL  | 190 | HTNVDGTEF  | 200 | GGSIYQVKNK | 210 | KLETAVNLAM | 220 | TAGNSNTRFG |
| 230 | IAAKYQIDPD  | 240 | ACFSAKVNS  | 250 | SLIGLGYTQT | 260 | LKPGIKLTLS | 270 | ALLDGKNVNA | 280 | GGHKLGLGLE | FQA |            |     |            |     |            |     |            |     |            |

### Computationally mapped potential isoform sequences<sup>i</sup>

There is 1 potential isoform mapped to this entry

**UniProt** BLAST Align Peptide search ID mapping SPARQL UniProtKB Advanced | List Search

Function Variant viewer 100 Feature viewer Genomic coordinates Publications External links History

BLAST Download Add Add a publication Entry feedback

### Function<sup>i</sup>

Forms a channel through the mitochondrial outer membrane and also the plasma membrane. The channel at the outer mitochondrial membrane allows diffusion of small hydrophilic molecules; in the plasma membrane it is involved in cell volume regulation and apoptosis. It adopts an open conformation at low or zero membrane potential and a closed conformation at potentials above 30-40 mV. The open state has a weak anion selectivity whereas the closed state is cation-selective (PubMed 11845315, PubMed 18755977, PubMed 20230784, PubMed 8420959). Binds various signaling molecules, including the sphingolipid ceramide, the phospholipid phosphatidylcholine, and the sterol cholesterol (PubMed 31015432). In depolarized mitochondria, acts downstream of PRKN and PINK1 to promote mitophagy or prevent apoptosis; polyubiquitination by PRKN promotes mitophagy, while monoubiquitination by PRKN decreases mitochondrial calcium influx which ultimately inhibits apoptosis (PubMed 32047033). May participate in the formation of the permeability transition pore complex (PTPC) responsible for the release of mitochondrial products that triggers apoptosis (PubMed 15033708, PubMed 25296756). May mediate ATP export from cells (PubMed 30061676). 9 Publications

### Activity regulation<sup>i</sup>

Inhibited by nitric oxide. 1 Publication

### Features

Showing features for site<sup>i</sup>, binding site<sup>i</sup>.

Download

1 20 40 60 80 100 120 140 160 180 200 220 240 260 280 283

UniProtKB - **UniprotKB** - Advanced | [Link](#) [Search](#) [Help](#)

Function  
Names & Taxonomy  
Subcellular Location  
Disease & Variants  
PTM/Processing  
Expression  
Interaction  
Structure  
Family & Domains  
Sequence  
Similar Proteins

Entry Variant viewer **100** Feature viewer Genomic coordinates Publications External links History

-- Select --

| Entry           | Variant viewer | Feature viewer | Genomic coordinates                               | Publications | External links | History |
|-----------------|----------------|----------------|---------------------------------------------------|--------------|----------------|---------|
| ► Transmembrane | 26-36          | Beta stranded  | <a href="#">15</a> <a href="#">2 Publications</a> | BLAST        | ⚙ Add          |         |
| ► Transmembrane | 39-47          | Beta stranded  | <a href="#">15</a> <a href="#">2 Publications</a> | BLAST        | ⚙ Add          |         |
| ► Transmembrane | 54-64          | Beta stranded  | <a href="#">15</a> <a href="#">2 Publications</a> | BLAST        | ⚙ Add          |         |
| ► Transmembrane | 69-76          | Beta stranded  | <a href="#">15</a> <a href="#">2 Publications</a> | BLAST        | ⚙ Add          |         |
| ► Transmembrane | 80-89          | Beta stranded  | <a href="#">15</a> <a href="#">2 Publications</a> | BLAST        | ⚙ Add          |         |
| ► Transmembrane | 95-104         | Beta stranded  | <a href="#">15</a> <a href="#">2 Publications</a> | BLAST        | ⚙ Add          |         |
| ► Transmembrane | 111-120        | Beta stranded  | <a href="#">15</a> <a href="#">2 Publications</a> | BLAST        | ⚙ Add          |         |
| ► Transmembrane | 123-130        | Beta stranded  | <a href="#">15</a> <a href="#">2 Publications</a> | BLAST        | ⚙ Add          |         |
| ► Transmembrane | 137-145        | Beta stranded  | <a href="#">15</a> <a href="#">2 Publications</a> | BLAST        | ⚙ Add          |         |
| ► Transmembrane | 150-158        | Beta stranded  | <a href="#">15</a> <a href="#">2 Publications</a> | BLAST        | ⚙ Add          |         |
| ► Transmembrane | 163-175        | Beta stranded  | <a href="#">15</a> <a href="#">2 Publications</a> | BLAST        | ⚙ Add          |         |
| ► Transmembrane | 178-185        | Beta stranded  | <a href="#">15</a> <a href="#">2 Publications</a> | BLAST        | ⚙ Add          |         |
| ► Transmembrane | 189-198        | Beta stranded  | <a href="#">15</a> <a href="#">2 Publications</a> | BLAST        | ⚙ Add          |         |
| ► Transmembrane | 202-211        | Beta stranded  | <a href="#">15</a> <a href="#">2 Publications</a> | BLAST        | ⚙ Add          |         |
| ► Transmembrane | 218-227        | Beta stranded  | <a href="#">15</a> <a href="#">2 Publications</a> | BLAST        | ⚙ Add          |         |
| ► Transmembrane | 231-238        | Beta stranded  | <a href="#">15</a> <a href="#">2 Publications</a> | BLAST        | ⚙ Add          |         |
| ► Transmembrane | 242-251        | Beta stranded  | <a href="#">15</a> <a href="#">2 Publications</a> | BLAST        | ⚙ Add          |         |
| ► Transmembrane | 254-263        | Beta stranded  | <a href="#">15</a> <a href="#">2 Publications</a> | BLAST        | ⚙ Add          |         |
| ► Transmembrane | 273-282        | Beta stranded  | <a href="#">15</a> <a href="#">2 Publications</a> | BLAST        | ⚙ Add          |         |

### Question 2

[illegible]

UniProt BLAST Align Peptide search ID mapping SPARQL UniRef - transcription factors

Advanced | List Search

**Taxonomy**  
Filter by taxonomy

**Clusters**  
100% (787)  
90% (564)  
50% (235)

### UniRef 1,586 results

BLAST Align Map IDs Download Add View: Cards Table Customize columns Share

| Cluster ID           | Cluster name                              | Types | Size       | Organisms                                                                                                                                                                       | Length | Identity  |
|----------------------|-------------------------------------------|-------|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-----------|
| UniRef100_A0A9P7SZP2 | Cluster: Transcription factors            | li    | 1 member   | Claviceps pusilla                                                                                                                                                               | 393    | UniRef100 |
| UniRef50_A0A9P7SZP2  | Cluster: Transcription factors            | li    | 1 member   | Claviceps pusilla                                                                                                                                                               | 393    | UniRef50  |
| UniRef50_A0A9P7SZP2  | Cluster: Transcription factors            | li    | 11 members | Claviceps pusilla<br>Coniochaeta sp. 272.1<br>Pseudovirgaria hyperparasitica<br>Cryphonectria parasitica EP155<br>Trichoderma parareesei (Filamentous fungus)<br>More organisms | 393    | UniRef50  |
| UniRef100_A0A0G2S.H4 | Cluster: AP10/EREBP Transcription Factors | li    | 1 member   | Salvia miltiorrhiza (Chinese sage)                                                                                                                                              | 471    | UniRef100 |
| UniRef100_A0A0G2S.N9 | Cluster: AP4/EREBP Transcription Factors  | li    | 1 member   | Salvia miltiorrhiza (Chinese sage)                                                                                                                                              | 461    | UniRef100 |
| UniRef100_A0A0M3R849 | Cluster: NAC transcription factors 3      | li    | 1 member   | Manihot esculenta (Cassava)                                                                                                                                                     | 581    | UniRef100 |
| UniRef100_A0A0M3R852 | Cluster: NAC transcription factors 17     | li    | 1 member   | Manihot esculenta (Cassava)                                                                                                                                                     | 358    | UniRef100 |
| UniRef100_A0A0M3R854 | Cluster: NAC transcription factors 27     | li    | 1 member   | Manihot esculenta (Cassava)                                                                                                                                                     | 311    | UniRef100 |
| UniRef100_A0A0M3R856 | Cluster: NAC transcription factors 37     | li    | 1 member   | Manihot esculenta (Cassava)                                                                                                                                                     | 217    | UniRef100 |
| UniRef100_A0A0M3R858 | Cluster: NAC transcription factors 47     | li    | 1 member   | Manihot esculenta (Cassava)                                                                                                                                                     | 293    | UniRef100 |
| UniRef100_A0A0M3R859 | Cluster: NAC transcription factors 53     | li    | 1 member   | Manihot esculenta (Cassava)                                                                                                                                                     | 385    | UniRef100 |
| UniRef100_A0A0M4FBP6 | Cluster: NAC transcription factors 15     | li    | 1 member   | Manihot esculenta (Cassava)                                                                                                                                                     | 366    | UniRef100 |

UniProt BLAST Align Peptide search ID mapping SPARQL UniRef - transcription factors

Advanced | List Search

**Taxonomy**  
Filter by taxonomy

**Clusters**  
50% (235)

### UniRef 235 results

BLAST Align Map IDs Download Add View: Cards Table Customize columns Share

| Cluster ID          | Cluster name                                       | Types | Size       | Organisms                                                                                                                                                                       | Length | Identity |
|---------------------|----------------------------------------------------|-------|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|----------|
| UniRef50_A0A9P7SZP2 | Cluster: Transcription factors                     | li    | 11 members | Claviceps pusilla<br>Coniochaeta sp. 272.1<br>Pseudovirgaria hyperparasitica<br>Cryphonectria parasitica EP155<br>Trichoderma parareesei (Filamentous fungus)<br>More organisms | 393    | UniRef50 |
| UniRef50_A0A9QINQ52 | Cluster: HOMEBOX PROTEIN TRANSCRIPTION FACTORS     | li    | 1 member   | Salix viminalis (Common osier)                                                                                                                                                  | 377    | UniRef50 |
| UniRef50_A0A9QIVHM2 | Cluster: HOMEBOX PROTEIN TRANSCRIPTION FACTORS     | li    | 1 member   | Salix purpurea (Purple osier willow)                                                                                                                                            | 111    | UniRef50 |
| UniRef50_A0A2P2Z288 | Cluster: NAC transcription factors 93              | li    | 1 member   | Rhizophora mucronata (Asiatic mangrove)                                                                                                                                         | 84     | UniRef50 |
| UniRef50_A0A2P2LA09 | Cluster: NAC transcription factors 5               | li    | 1 member   | Rhizophora mucronata (Asiatic mangrove)                                                                                                                                         | 74     | UniRef50 |
| UniRef50_A0A2P2NJ50 | Cluster: NAC transcription factors 7               | li    | 1 member   | Rhizophora mucronata (Asiatic mangrove)                                                                                                                                         | 59     | UniRef50 |
| UniRef50_A0A2P2PUA9 | Cluster: NAC transcription factors 88              | li    | 1 member   | Rhizophora mucronata (Asiatic mangrove)                                                                                                                                         | 21     | UniRef50 |
| UniRef50_A0A9Q0TUE3 | Cluster: HOMEBOX PROTEIN TRANSCRIPTION FACTORS     | li    | 1 member   | Salix purpurea (Purple osier willow)                                                                                                                                            | 201    | UniRef50 |
| UniRef50_A0A9QIWL8E | Cluster: HOMEBOX PROTEIN TRANSCRIPTION FACTORS     | li    | 1 member   | Salix koriyanagi                                                                                                                                                                | 112    | UniRef50 |
| UniRef50_Q2L0L5     | Cluster: Virulence factors transcription regulator | li    | 31 members | Bordetella avium (strain 197N)<br>Bordetella avium<br>Bordetella hinzii<br>Bordetella pseudohinzii<br>Bordetella holmesii<br>More organisms                                     | 287    | UniRef50 |

For 50% the Cluster count is – 235

D2 41

|     | A                    | B         | C    | D            | E   | F   | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W |
|-----|----------------------|-----------|------|--------------|-----|-----|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 220 | UniRef50_Cluster: Si | UniProtKB | 4    | Culex quin   | 286 | 0.5 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 221 | UniRef50_Cluster: Si | UniProtKB | 1    | Culex quin   | 311 | 0.5 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 222 | UniRef50_Cluster: Si | UniProtKB | 1    | Culex quin   | 145 | 0.5 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 223 | UniRef50_Cluster: M  | UniProtKB | 6    | Harpegnat    | 199 | 0.5 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 224 | UniRef50_Cluster: M  | UniProtKB | 1    | Daphnia pi   | 361 | 0.5 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 225 | UniRef50_Cluster: M  | UniProtKB | 1    | Solenopsis   | 92  | 0.5 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 226 | UniRef50_Cluster: M  | UniProtKB | 1    | Macaca fa    | 45  | 0.5 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 227 | UniRef50_Cluster: Vi | UniProtKB | 2432 | Bordetella   | 209 | 0.5 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 228 | UniRef50_Cluster: G1 | UniProtKB | 20   | Saccharon    | 629 | 0.5 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 229 | UniRef50_Cluster: Vi | UniProtKB | 31   | Bordetella   | 207 | 0.5 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 230 | UniRef50_Cluster: Pu | UniProtKB | 1    | Rattus nor   | 108 | 0.5 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 231 | UniRef50_Cluster: M  | UniProtKB | 1    | Capitella t. | 308 | 0.5 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 232 | UniRef50_Cluster: M  | UniProtKB | 1    | Capitella t. | 140 | 0.5 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 233 | UniRef50_Cluster: M  | UniProtKB | 1    | Megaselia    | 227 | 0.5 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 234 | UniRef50_Cluster: M  | UniProtKB | 1    | Megaselia    | 80  | 0.5 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 235 | UniRef50_Cluster: Pu | UniProtKB | 1    | Corethrelli  | 244 | 0.5 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 236 | UniRef50_Cluster: M  | UniProtKB | 2    | Necator ai   | 141 | 0.5 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 237 |                      |           |      |              |     |     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 238 |                      |           |      |              |     |     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 239 |                      |           |      |              |     |     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 240 |                      |           |      |              |     |     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 241 |                      |           |      |              |     |     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 242 |                      |           |      |              |     |     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 243 |                      |           |      |              |     |     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 244 |                      |           |      |              |     |     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |

Sheet0

Ready Accessibility: Good to go

Average: 18.51914894 Count: 235 Sum: 4352

|     |                                |      |             |     |     |
|-----|--------------------------------|------|-------------|-----|-----|
| 225 | UniRef50_Cluster: M UniProtKB  | 1    | Solenopsis  | 92  | 0.5 |
| 226 | UniRef50_Cluster: M UniProtKB  | 1    | Macaca fa   | 45  | 0.5 |
| 227 | UniRef50_Cluster: Vi UniProtKB | 2432 | Bordetella  | 209 | 0.5 |
| 228 | UniRef50_Cluster: G UniProtKB  | 20   | Saccharom   | 629 | 0.5 |
| 229 | UniRef50_Cluster: Vi UniProtKB | 31   | Bordetella  | 207 | 0.5 |
| 230 | UniRef50_Cluster: Pu UniProtKB | 1    | Rattus nor  | 108 | 0.5 |
| 231 | UniRef50_Cluster: M UniProtKB  | 1    | Capitella t | 308 | 0.5 |
| 232 | UniRef50_Cluster: M UniProtKB  | 1    | Capitella t | 140 | 0.5 |
| 233 | UniRef50_Cluster: M UniProtKB  | 1    | Megaselia   | 227 | 0.5 |
| 234 | UniRef50_Cluster: M UniProtKB  | 1    | Megaselia   | 80  | 0.5 |
| 235 | UniRef50_Cluster: Pu UniProtKB | 1    | Corethrelli | 244 | 0.5 |
| 236 | UniRef50_Cluster: M UniProtKB  | 2    | Necator ai  | 141 | 0.5 |
| 237 |                                |      |             |     |     |
| 238 |                                |      |             |     |     |
| 239 |                                |      |             |     |     |
| 240 |                                |      |             |     |     |
| 241 |                                |      |             |     |     |
| 242 |                                |      |             |     |     |
| 243 |                                |      |             |     |     |
| 244 |                                |      |             |     |     |
| 245 |                                |      |             |     |     |

Sheet0

Ready Accessibility: Good to go

Average: 18.51914894 Count: 235 Sum: 4352

The sum of sequences for 50%- 4352

Similarly, for 90%, it is -2110

### Question 3

| Cluster ID | Cluster Na | Types       | Size | Organisms | Length | Identity |
|------------|------------|-------------|------|-----------|--------|----------|
| 1          | UniRef100  | Cluster: Re | 1    | Homo sap  | 121    | 1.0      |
| 2          | UniRef100  | Cluster: Cy | 1    | Homo sap  | 380    | 1.0      |
| 3          | UniRef100  | Cluster: Nu | 1    | Homo sap  | 603    | 1.0      |
| 4          | UniRef100  | Cluster: A1 | 1    | Homo sap  | 226    | 1.0      |
| 5          | UniRef100  | Cluster: Nu | 1    | Homo sap  | 547    | 1.0      |
| 6          | UniRef100  | Cluster: A1 | 1    | Homo sap  | 226    | 1.0      |
| 7          | UniRef100  | Cluster: Nu | 1    | Homo sap  | 318    | 1.0      |
| 8          | UniRef100  | Cluster: A1 | 1    | Homo sap  | 226    | 1.0      |
| 9          | UniRef100  | Cluster: Nu | 1    | Homo sap  | 603    | 1.0      |
| 10         | UniRef100  | Cluster: Nu | 1    | Homo sap  | 603    | 1.0      |
| 11         | UniRef100  | Cluster: Nu | 1    | Homo sap  | 603    | 1.0      |
| 12         | UniRef100  | Cluster: Nu | 1    | Homo sap  | 603    | 1.0      |
| 13         | UniRef100  | Cluster: A1 | 1    | Homo sap  | 226    | 1.0      |
| 14         | UniRef100  | Cluster: Nu | 1    | Homo sap  | 603    | 1.0      |
| 15         | UniRef100  | Cluster: Nu | 1    | Homo sap  | 603    | 1.0      |
| 16         | UniRef100  | Cluster: Cy | 1    | Homo sap  | 380    | 1.0      |
| 17         | UniRef100  | Cluster: Nu | 1    | Homo sap  | 347    | 1.0      |
| 18         | UniRef100  | Cluster: Nu | 1    | Homo sap  | 174    | 1.0      |
| 19         | UniRef100  | Cluster: Nu | 1    | Homo sap  | 603    | 1.0      |
| 20         | UniRef100  | Cluster: A1 | 1    | Homo sap  | 226    | 1.0      |
| 21         | UniRef100  | Cluster: Nu | 1    | Homo sap  | 603    | 1.0      |
| 22         | UniRef100  | Cluster: Nu | 1    | Homo sap  | 603    | 1.0      |
| 23         | UniRef100  | Cluster: Cy | 1    | Homo sap  | 513    | 1.0      |
| 24         | UniRef100  | Cluster: Cy | 1    | Homo sap  | 513    | 1.0      |
| 25         | UniRef100  | Cluster: A1 | 1    | Homo sap  | 226    | 1.0      |

490357

Sheet0

Ready Accessibility: Investigate

100%

Excel spreadsheet showing UniProtKB clusters and their associated protein counts and scores. The data is organized in columns A through W. The first column (A) contains UniRef90 IDs, the second column (B) contains UniRef90 cluster names, the third column (C) contains UniProtKB IDs, the fourth column (D) contains protein counts, the fifth column (E) contains protein names, and the sixth column (F) contains scores.

| UniRef90 ID | UniRef90 Cluster                | UniProtKB ID  | Protein Count | Protein Name | Score |
|-------------|---------------------------------|---------------|---------------|--------------|-------|
| 2554        | UniRef90_Cluster: N1 UniProtKB  | 1 Rhizoctonia | 928           |              | 0.9   |
| 2555        | UniRef90_Cluster: H4 UniProtKB  | 1 Rhizoctonia | 602           |              | 0.9   |
| 2556        | UniRef90_Cluster: N4 UniProtKB  | 1 Rhizoctonia | 1044          |              | 0.9   |
| 2557        | UniRef90_Cluster: N4 UniProtKB  | 1 Rhizoctonia | 1190          |              | 0.9   |
| 2558        | UniRef90_Cluster: N4 UniProtKB  | 1 Rhizoctonia | 1592          |              | 0.9   |
| 2559        | UniRef90_Cluster: S0 UniProtKB  | 1 Rhizoctonia | 594           |              | 0.9   |
| 2560        | UniRef90_Cluster: Cc UniProtKB  | 3 Homo sapi   | 73            |              | 0.9   |
| 2561        | UniRef90_Cluster: L11 UniProtKB | 2 Homo sapi   | 209           |              | 0.9   |
| 2562        | UniRef90_Cluster: Cc UniProtKB  | 6 Homo sapi   | 175           |              | 0.9   |
| 2563        | UniRef90_Cluster: Tr UniProtKB  | 3 Homo sapi   | 122           |              | 0.9   |
| 2564        | UniRef90_Cluster: Fa UniProtKB  | 2 Homo sapi   | 225           |              | 0.9   |
| 2565        | UniRef90_Cluster: H1 UniProtKB  | 11 Homo sapi  | 353           |              | 0.9   |
| 2566        | UniRef90_Cluster: M UniProtKB   | 9 Homo sapi   | 89            |              | 0.9   |
| 2567        | UniRef90_Cluster: S0 UniProtKB  | 1 Homo sapi   | 25            |              | 0.9   |
| 2568        | UniRef90_Cluster: 2 UniProtKB   | 1 Homo sapi   | 57            |              | 0.9   |
| 2569        | UniRef90_Cluster: Fr UniProtKB  | 1 Homo sapi   | 594           |              | 0.9   |
| 2570        | UniRef90_Cluster: M UniProtKB   | 7 Homo sapi   | 181           |              | 0.9   |
| 2571        | UniRef90_Cluster: B1 UniProtKB  | 1 Homo sapi   | 149           |              | 0.9   |
| 2572        | UniRef90_Cluster: 2 UniProtKB   | 1 Homo sapi   | 68            |              | 0.9   |
| 2573        | UniRef90_Cluster: M UniProtKB   | 23 Homo sapi  | 206           |              | 0.9   |
| 2574        | UniRef90_Cluster: Tr UniProtKB  | 1 Homo sapi   | 21            |              | 0.9   |
| 2575        | UniRef90_Cluster: Tr UniProtKB  | 1 Homo sapi   | 98            |              | 0.9   |
| 2576        | UniRef90_Cluster: N4 UniProtKB  | 1 Homo sapi   | 100           |              | 0.9   |
| 2577        | UniRef90_Cluster: Cy UniProtKB  | 4 Homo sapi   | 119           |              | 0.9   |
| 2578        | UniRef90_Cluster: Tr UniProtKB  | 1 Homo sapi   | 113           |              | 0.9   |

Excel spreadsheet showing UniProtKB clusters and their associated protein counts and scores. The data is organized in columns A through W. The first column (A) contains UniRef50 IDs, the second column (B) contains UniRef50 cluster names, the third column (C) contains UniProtKB IDs, the fourth column (D) contains protein counts, the fifth column (E) contains protein names, and the sixth column (F) contains scores. The formula bar shows the formula for cell E54157: `=SUM(D2:D54155)`.

| UniRef50 ID | UniRef50 Cluster               | UniProtKB ID  | Protein Count | Protein Name | Score |
|-------------|--------------------------------|---------------|---------------|--------------|-------|
| 54137       | UniRef50_Cluster: R1 UniProtKB | 10 Homo sapi  | 129           |              | 0.5   |
| 54138       | UniRef50_Cluster: P1 UniProtKB | 111 Homo sapi | 127           |              | 0.5   |
| 54139       | UniRef50_Cluster: SC UniProtKB | 13 Homo sapi  | 74            |              | 0.5   |
| 54140       | UniRef50_Cluster: N1 UniProtKB | 1 Homo sapi   | 155           |              | 0.5   |
| 54141       | UniRef50_Cluster: In UniProtKB | 1 Homo sapi   | 324           |              | 0.5   |
| 54142       | UniRef50_Cluster: M UniProtKB  | 3 Homo sapi   | 160           |              | 0.5   |
| 54143       | UniRef50_Cluster: D1 UniProtKB | 7 Homo sapi   | 881           |              | 0.5   |
| 54144       | UniRef50_Cluster: Pr UniProtKB | 4 Homo sapi   | 124           |              | 0.5   |
| 54145       | UniRef50_Cluster: M UniProtKB  | 3 Homo sapi   | 46            |              | 0.5   |
| 54146       | UniRef50_Cluster: Zi UniProtKB | 3 Homo sapi   | 41            |              | 0.5   |
| 54147       | UniRef50_Cluster: M UniProtKB  | 15 Homo sapi  | 241           |              | 0.5   |
| 54148       | UniRef50_Cluster: N1 UniProtKB | 6 Homo sapi   | 53            |              | 0.5   |
| 54149       | UniRef50_Cluster: AF UniProtKB | 2 Homo sapi   | 61            |              | 0.5   |
| 54150       | UniRef50_Cluster: Pr UniProtKB | 3 Homo sapi   | 85            |              | 0.5   |
| 54151       | UniRef50_Cluster: Pr UniProtKB | 3 Homo sapi   | 171           |              | 0.5   |
| 54152       | UniRef50_Cluster: L UniProtKB  | 15 Homo sapi  | 127           |              | 0.5   |
| 54153       | UniRef50_Cluster: tR UniProtKB | 7 Homo sapi   | 212           |              | 0.5   |
| 54154       | UniRef50_Cluster: F UniProtKB  | 6 Homo sapi   | 298           |              | 0.5   |
| 54155       | UniRef50_Cluster: Cc UniProtKB | 10 Homo sapi  | 218           |              | 0.5   |

From the above images:

100% - 490357

90% - 2062761

50%-8440805

Question 4

UniProt BLAST Align Peptide search ID mapping SPARQL UniProtKB (taxonomy\_id:10090) AND (reviewed:true)

Status  
Reviewed (Swiss-Prot) (17,208)

Popular organisms  
Mouse (17,201)

Taxonomy  
10090

Group by  
Taxonomy  
Keywords  
Gene Ontology  
Enzyme Class

Proteins with  
3D structure (2,351)  
Active site (2,234)  
Activity regulation (1,285)  
Allergen (1)  
Alternative products (isoforms) (4,988)  
More items

Protein existence  
Protein level (13,129)  
Transcript level (3,644)  
Homology (364)  
Predicted (63)

### UniProtKB 17,208 results

or restrict search to "10090" to exclude lower taxonomic ranks

BLAST Align Map IDs Download Add View: Cards Table Customize columns Share

| Entry      | Entry Name   | Protein Names                                              | Gene Names                 | Organism             | Length   |
|------------|--------------|------------------------------------------------------------|----------------------------|----------------------|----------|
| A0A5F8MPU3 | CTSR_MOUSE   | Cation channel sperm-associated targeting subunit tau[...] | Czcd6, Abzot11, CATSPERT   | Mus musculus (Mouse) | 2,282 AA |
| A0A5K7RLP0 | MEIOS_MOUSE  | Meiosis initiator protein                                  | Meiosin, Bhm1              | Mus musculus (Mouse) | 589 AA   |
| A2A891     | CMTA1_MOUSE  | Calmodulin-binding transcription activator 1               | Canta1, Kiaa0833           | Mus musculus (Mouse) | 1,682 AA |
| A2A9F4     | KDF1_MOUSE   | Keratinocyte differentiation factor 1                      | Kdf1                       | Mus musculus (Mouse) | 397 AA   |
| A2AC93     | DNAI2_MOUSE  | Dynein axonemal intermediate chain 2[...]                  | Dnai2, Dnaic2              | Mus musculus (Mouse) | 623 AA   |
| A2AF47     | DOC1_MOUSE   | Dedicator of cytokinesis protein 11[...]                   | Dock11, Zic2               | Mus musculus (Mouse) | 2,073 AA |
| A2AGT5     | CKAP5_MOUSE  | Cytoskeleton-associated protein 5                          | Ckap5                      | Mus musculus (Mouse) | 2,032 AA |
| A2AIW0     | ENTR1_MOUSE  | Endosome-associated-trafficking regulator 1[...]           | Entr1, Sdcccag3            | Mus musculus (Mouse) | 432 AA   |
| A2AJB7     | LCN5_MOUSE   | Epilidymal-specific lipocalin 5[...]                       | Lcn5, Mep10                | Mus musculus (Mouse) | 192 AA   |
| A2AJK6     | CHD7_MOUSE   | Chromodomain-helicase-DNA binding protein 7[...]           | Chd7                       | Mus musculus (Mouse) | 2,986 AA |
| A2AJX4     | MALR1_MOUSE  | MAM and LDL receptor class A domain-containing protein 1   | Malr1, Dier1, Gm13318      | Mus musculus (Mouse) | 2,123 AA |
| A2AKQ0     | S3SD1_MOUSE  | Nucleotide sugar transporter SLC35D1[...]                  | Slc35d1                    | Mus musculus (Mouse) | 355 AA   |
| A2AL55     | RPQP1_MOUSE  | Rap1 GTPase-activating protein 1[...]                      | Rap1gap, Kiaa0474, Rap1ga1 | Mus musculus (Mouse) | 663 AA   |
| A2APB8     | TPX2_MOUSE   | Targeting protein for Klp2                                 | Tpx2                       | Mus musculus (Mouse) | 745 AA   |
| A2APC3     | TTL9_MOUSE   | Probable tubulin polyglutamylation TTL9[...]               | Ttl9                       | Mus musculus (Mouse) | 461 AA   |
| A2AQ07     | TUBB1_MOUSE  | Tubulin beta-1 chain                                       | Tubb1                      | Mus musculus (Mouse) | 451 AA   |
| A2AU37     | RD21L_MOUSE  | Double-strand-break repair protein rad21-like protein 1    | Ra221l1, Gm14160, Ra221l   | Mus musculus (Mouse) | 552 AA   |
| A2AWP0     | BIRC7_MOUSE  | Baculoviral IAP repeat-containing protein 7[...]           | Birc7, Livin               | Mus musculus (Mouse) | 285 AA   |
| A2AX52     | COL6A4_MOUSE | Collagen alpha 4(VI) chain                                 | Col6a4, Dvra               | Mus musculus (Mouse) | 2,369 AA |

https://www.uniprot.org/uniprotkb?query=%28taxonomy\_id%3A10090%29+AND+%28structure\_3d%3Atrue%29

UniProt BLAST Align Peptide search ID mapping SPARQL UniProtKB (taxonomy\_id:10090) AND (structure\_3d:true)

Status  
Reviewed (Swiss-Prot) (2,351)  
Unreviewed (TrEMBL) (168)

Popular organisms  
Mouse (2,519)

Taxonomy  
10090

Group by  
Taxonomy  
Keywords  
Gene Ontology  
Enzyme Class

Proteins with  
3D structure (2,519)

### UniProtKB 2,519 results

or restrict search to "10090" to exclude lower taxonomic ranks

BLAST Align Map IDs Download Add View: Cards Table Customize columns Share

| Entry  | Entry Name  | Protein Names                                                | Gene Names        | Organism             | Length   |
|--------|-------------|--------------------------------------------------------------|-------------------|----------------------|----------|
| A2CG49 | KALRN_MOUSE | Kalirin[...]                                                 | Kalrn             | Mus musculus (Mouse) | 2,964 AA |
| B2RX54 | PLXB2_MOUSE | Plexin-B2                                                    | Plxbn2            | Mus musculus (Mouse) | 1,842 AA |
| B5KM66 | SYCE3_MOUSE | Synaptonemal complex central element protein 3[...]          | Syce3, Tseg2      | Mus musculus (Mouse) | 88 AA    |
| E9Q9F6 | CTSRD_MOUSE | Cation channel sperm-associated auxiliary subunit delta[...] | Catsperd, Tmem146 | Mus musculus (Mouse) | 805 AA   |
| E9Q9W7 | PDZD7_MOUSE | PDZ domain-containing protein 7                              | Pdzd7             | Mus musculus (Mouse) | 1,021 AA |
| J3QMY9 | TO6BL_MOUSE | Type 2 DNA topoisomerase 6 subunit B-like[...]               | Top6bl, Gm960     | Mus musculus (Mouse) | 579 AA   |
| O08762 | NETR_MOUSE  | Neurotrophin[...]                                            | Prss12, Bssp3     | Mus musculus (Mouse) | 761 AA   |

We'd like to inform you that we have updated our Privacy Notice to comply with Europe's new General Data Protection Regulation (GDPR) that applies since 25 May 2018. [Accept](#)

2519 structures were associated with PDB ( 3D structures)

## Question 5

UniProt BLAST Align Peptide search ID mapping SPARQL **Test results**

Taxonomy  
Filter by taxonomy

Clusters  
90% (10)

### ID mapping 10 results found for UniProtKB\_AC-ID → UniRef90

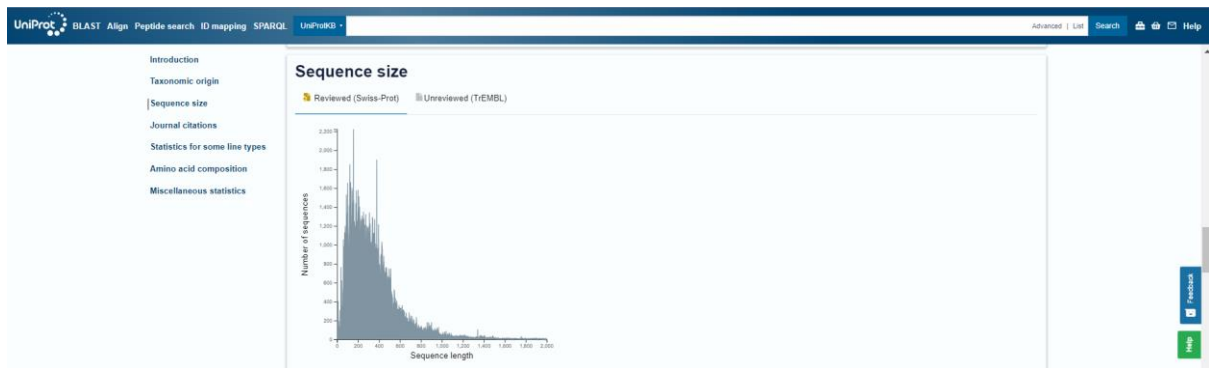
Overview Input Parameters API Request

BLAST Align Map IDs Download Add View: Cards Table Customize columns Resubmit Share

10 IDs were mapped to 10 results

| From   | Cluster ID      | Cluster name                                                     | Types      | Size | Organisms                                                                                                                                                                      | Length | Identity |
|--------|-----------------|------------------------------------------------------------------|------------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|----------|
| A2CG49 | UniRef90_A2CG49 | Cluster: Kalirin                                                 | 1 member   | 1    | Mus musculus (Mouse)                                                                                                                                                           | 2,964  | UniRef90 |
| B2RX54 | UniRef90_B2RX54 | Cluster: Plexin-B2                                               | 5 members  | 5    | Mus musculus (Mouse)<br>Mus caroli (Ryukyu mouse)<br>Mus spicilegus (Steppe mouse)                                                                                             | 1,842  | UniRef90 |
| B5KM66 | UniRef90_B5KM66 | Cluster: Synaptonemal complex central element protein 3          | 57 members | 57   | Mus musculus (Mouse)<br>Mus caroli (Ryukyu mouse)<br>Mus spicilegus (Steppe mouse)<br>Camelus ferus (Wild bactrian camel)<br>Camelus dromedarius (Dromedary)<br>More organisms | 88     | UniRef90 |
| E9Q9F6 | UniRef90_E9Q9F6 | Cluster: Cation channel sperm-associated auxiliary subunit delta | 17 members | 17   | Mus musculus (Mouse)<br>Mus spicilegus (Steppe mouse)<br>Mus caroli (Ryukyu mouse)                                                                                             | 805    | UniRef90 |
| E9Q9W7 | UniRef90_E9Q9W7 | Cluster: PDZ domain-containing protein 7                         | 20 members | 20   | Mus musculus (Mouse)<br>Neotoma lepida (Desert woodrat)                                                                                                                        | 1,021  | UniRef90 |

## Question 6

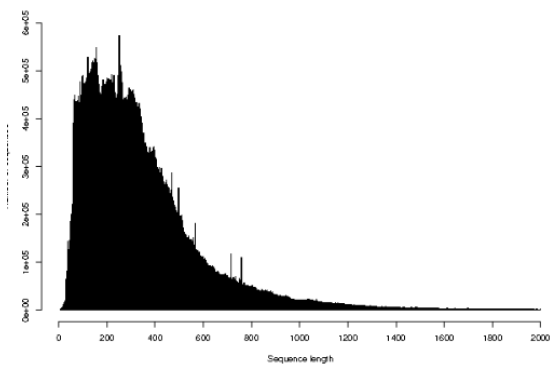


### 3. SEQUENCE SIZE

Repartition of the sequences by size (excluding fragments)

| From | To   | Number   | From  | To   | Number  |
|------|------|----------|-------|------|---------|
| 1-   | 50   | 2632209  | 1001- | 1100 | 1975924 |
| 51-  | 100  | 19670652 | 1101- | 1200 | 1399742 |
| 101- | 150  | 24534395 | 1201- | 1300 | 962640  |
| 151- | 200  | 23647606 | 1301- | 1400 | 667576  |
| 201- | 250  | 23258869 | 1401- | 1500 | 520074  |
| 251- | 300  | 23002062 | 1501- | 1600 | 392308  |
| 301- | 350  | 21192409 | 1601- | 1700 | 302782  |
| 351- | 400  | 16678825 | 1701- | 1800 | 238692  |
| 401- | 450  | 14161814 | 1801- | 1900 | 212549  |
| 451- | 500  | 11413729 | 1901- | 2000 | 181573  |
| 501- | 550  | 8239205  | 2001- | 2100 | 139701  |
| 551- | 600  | 6235574  | 2101- | 2200 | 123722  |
| 601- | 650  | 4694789  | 2201- | 2300 | 103755  |
| 651- | 700  | 3738600  | 2301- | 2400 | 83070   |
| 701- | 750  | 3175497  | 2401- | 2500 | 71972   |
| 751- | 800  | 2679062  | >2500 |      | 530559  |
| 801- | 850  | 2160451  |       |      |         |
| 851- | 900  | 1839023  |       |      |         |
| 901- | 950  | 1455535  |       |      |         |
| 951- | 1000 | 1153704  |       |      |         |

Length distribution of the sequences



The average sequence length in UniProtKB/TrEMBL is 351 amino acids.

The shortest sequence is A0A0G2JLF7\_HUMAN: 7 amino acids.

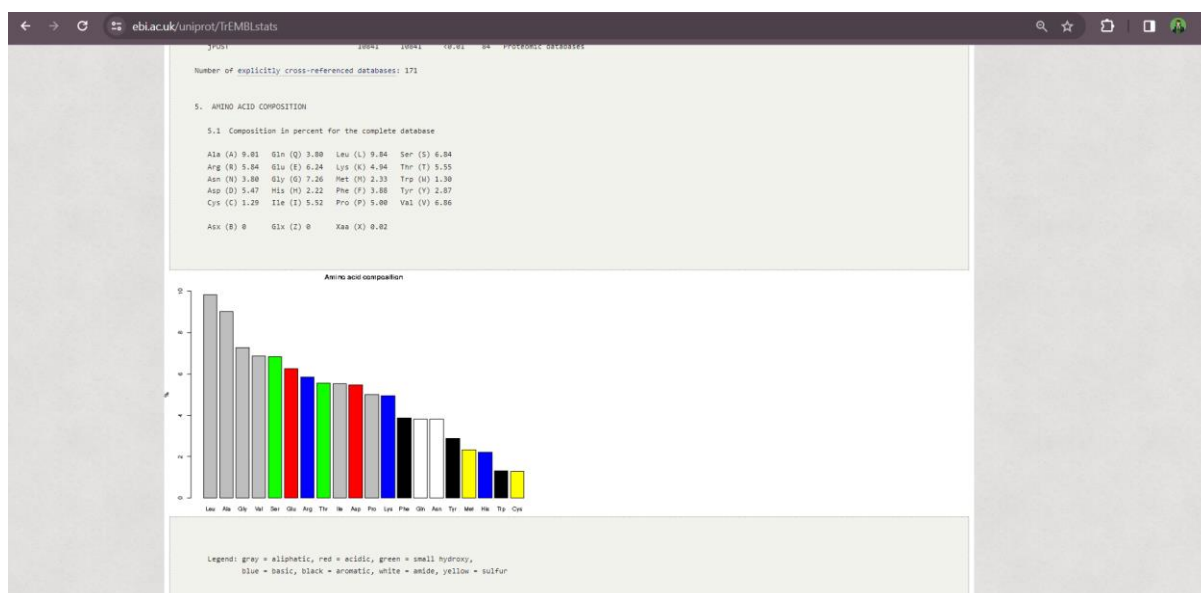
The longest sequence is A0A5A9P0L4\_9TELE: 45354 amino acids.

#### 4. STATISTICS FOR SOME LINE TYPES

The following table summarizes the total number of some UniProtKB/TrEMBL lines, as well as the number of entries with at least one such line, and the frequency of the lines.

| Line type / subtype           | Total number | Number of entries | Average per entry |
|-------------------------------|--------------|-------------------|-------------------|
| References (RL)               | 292116799    |                   | 1.17              |
| Submitted to EMBL/GenBank/DBJ | 193316258    | 178389382         | 0.77              |
| Journal                       | 80672071     | 76785321          | 0.32              |
| Submitted to other databases  | 18005494     | 17686805          | 0.07              |
| Thesis                        | 102563       | 102504            | <0.01             |
| Book citation                 | 20413        | 20345             | <0.01             |

Total number of distinct authors cited in UniProtKB/TrEMBL: 907189



c)

#### 5.1 Composition in percent for the complete database

|              |              |              |              |
|--------------|--------------|--------------|--------------|
| Ala (A) 9.01 | Gln (Q) 3.80 | Leu (L) 9.84 | Ser (S) 6.84 |
| Arg (R) 5.84 | Glu (E) 6.24 | Lys (K) 4.94 | Thr (T) 5.55 |
| Asn (N) 3.80 | Gly (G) 7.26 | Met (M) 2.33 | Trp (W) 1.30 |
| Asp (D) 5.47 | His (H) 2.22 | Phe (F) 3.88 | Tyr (Y) 2.87 |
| Cys (C) 1.29 | Ile (I) 5.52 | Pro (P) 5.00 | Val (V) 6.86 |
| Asx (B) 0    | Glx (Z) 0    | Xaa (X) 0.02 |              |